

PLPA6820_CodingChallenge4

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Question 1a: Explain the following concepts

- YAML Header: gives information on the file title, author, data, and output. My YAML header has my name, shows that I created my R markdown file today, and that it will render as a word document.
- Literate programming: allows individuals to understand the purpose of your code and different steps by combining the programming language with daily language.

Question 1b: Read the data using a relative file path

```
Mycotoxin <- read.csv("/Users/brias/OneDrive/Documents/briarepository/MycotoxinData.csv", na.strings = "
```

Question 1c: Make a separate code chunk for the figures plotting the data

Plot 1

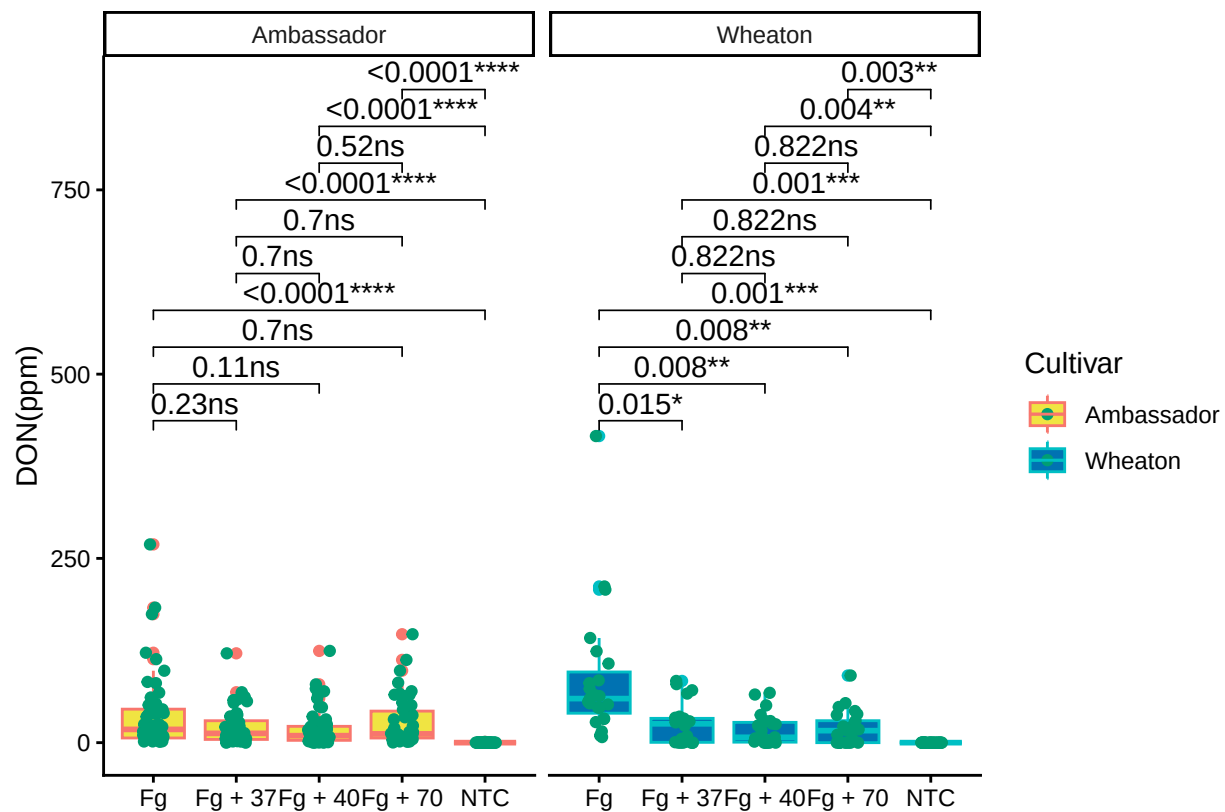
```
library(ggplot2)
library(ggpubr)
Plot_1 <- ggplot(Mycotoxin, aes(x = Treatment, y = DON, color = Cultivar, fill = Cultivar)) +
  geom_boxplot() +
  geom_point(position=position_jitterdodge(dodge.width = 0.6), color = "#009E73") + # coloring points,
  scale_fill_manual(values=c("#F0E442", "#0072B2")) + # filling and coloring boxplots
  xlab("") + # labeling x-axis
  ylab("DON(ppm)") + # labeling y-axis
  theme_classic() + # classic theme
  facet_wrap(~Cultivar) # faceted by Cultivar
Mycotoxin$Treatment<- as.factor(Mycotoxin$Treatment)
Mycotoxin$Treatment<- factor(Mycotoxin$Treatment, levels = c("NTC", "Fg", "Fg + 37", "Fg + 40", "Fg + 72"))
```

```
Plot_1t <- Plot_1 +
  geom_pwc(aes(group = Treatment), method = "t_test", label = "{p.adj.format}{p.adj.signif}")
plot(Plot_1t)
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## ('stat_pwc()').
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Plot 2

```
Plot_2 <- ggplot(Mycotoxin, aes(x = Treatment, y = X15ADON, color = Cultivar, fill = Cultivar)) +
  geom_boxplot() +
  geom_point(position=position_jitterdodge(dodge.width = 0.6), color = "#009E73") + # coloring points,
  scale_fill_manual(values=c("#F0E442", "#0072B2")) + # filling and coloring boxplots
  xlab("") + # labeling x-axis
  ylab("15ADON") + # labeling y-axis
  theme_classic() + # classic theme
```

```

  facet_wrap(~Cultivar) # faceted by Cultivar
Plot_2t <- Plot_2 +
  geom_pwc(aes(group = Treatment), method = "t_test", label = "{p.adj.format}{p.adj.signif}")
plot(Plot_2t)

```

```

## Warning: Removed 10 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

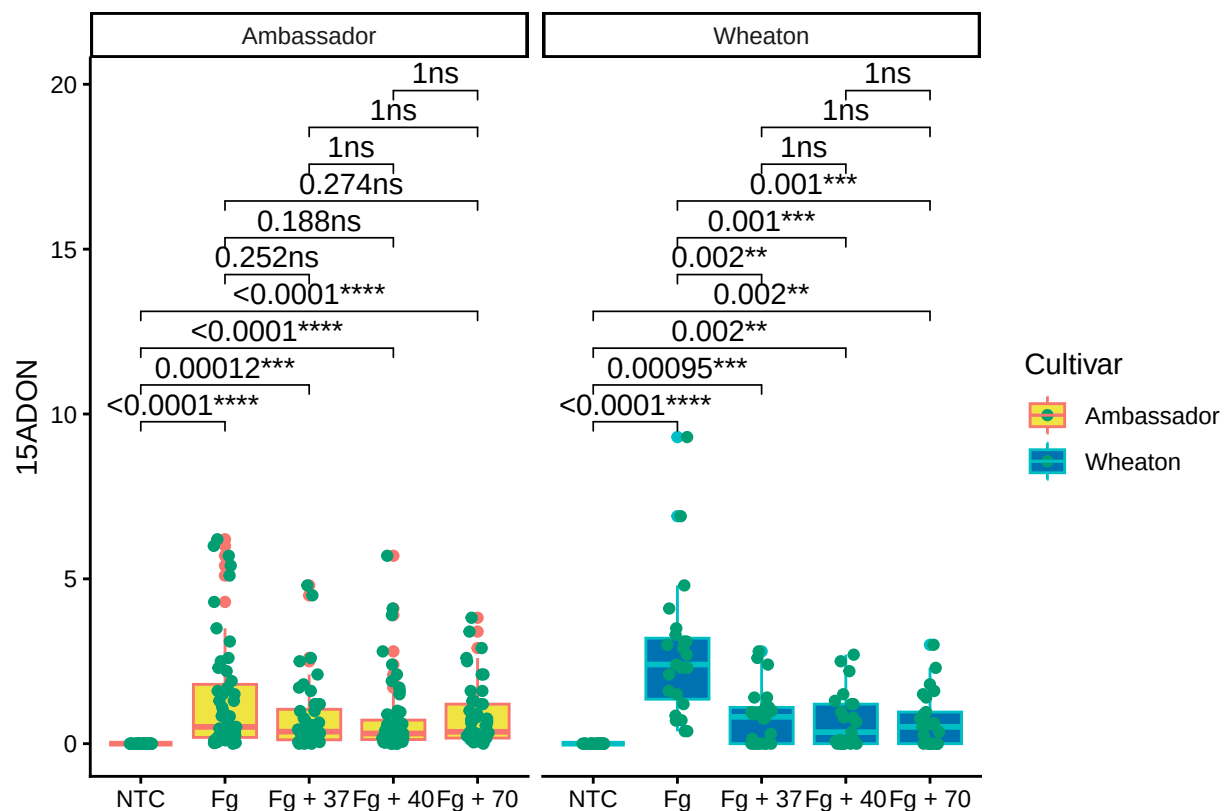
## Warning: Removed 10 rows containing non-finite outside the scale range
## ('stat_pwc()').

```

```

## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').

```



Plot 3

```

Plot_3 <- ggplot(Mycotoxin, aes(x = Treatment, y = MassperSeed_mg, color = Cultivar, fill = Cultivar)) +
  geom_boxplot() +
  geom_point(position=position_jitterdodge(dodge.width = 0.6), color = "#009E73") + # coloring points,
  scale_fill_manual(values=c("#F0E442", "#0072B2")) + # filling and coloring boxplots
  xlab("") + # labeling x-axis
  ylab("Seed Mass (mg)") + # labeling y-axis

```

```

theme_classic() + # classic theme
facet_wrap(~Cultivar) # faceted by Cultivar
Plot_3t <- Plot_3 +
  geom_pwc(aes(group = Treatment), method = "t_test", label = "{p.adj.format}{p.adj.signif}")
plot(Plot_3t)

```

```

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

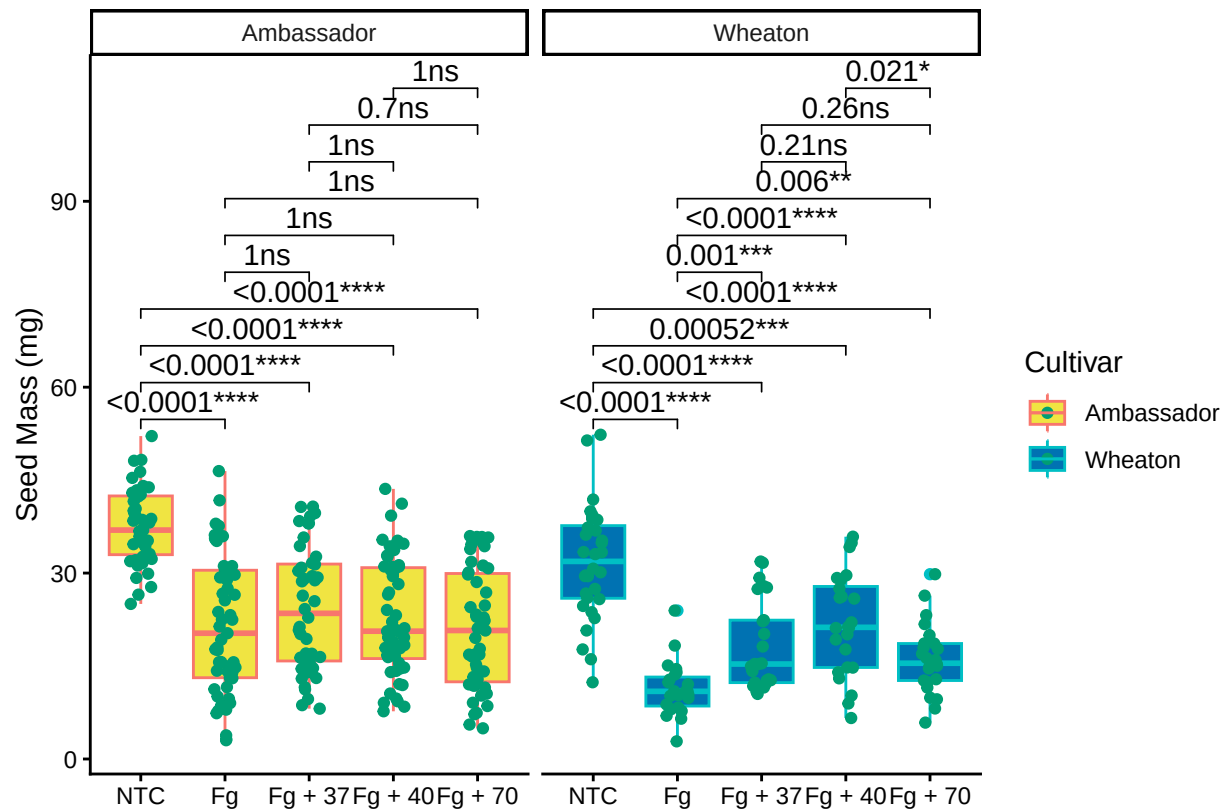
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_pwc()').

```

```

## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').

```



Combined Plots

```

combined_plot <- ggarrange( # combining the plots
  Plot_1t,
  Plot_2t,
  Plot_3t,
  ncol = 3,

```

```
nrow=1,  
labels = "auto", # labeling the subplots as a, b, and c  
common.legend = TRUE # RESPONSE TO QUESTION 4: common.legend gives all the plots one legend when thei  
)
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range  
## ('stat_pwc()').
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range  
## ('stat_pwc()').
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range  
## ('stat_pwc()').
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range  
## ('stat_pwc()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

```
plot(combined_plot)
```

Cultivar Ambassador Wheaton

